SFWRENG-3O03 Assignment 4

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Contents

1) LTI Systems

Given the two impulse responses:

$$h_1(n) = \begin{cases} 1, & \text{if } n \text{ is even and } n \geq 0 \\ 0, & \text{if } n \text{ is odd or } n < 0 \end{cases}$$

$$h_2(n) = \delta(n) - \delta(n-1) + \delta(n-2) - \delta(n-3)$$

1.a) Find the difference equation representation of the system with impulse response $h_1(n)$.

Solution:

This one is a little tricky. My first instinct was to simply make the difference equation of the system $y(n) = \sum_{k=0}^{\infty}(\delta(n-k))$ with only even values of k, however the difference equation of a system cannot be an infinite sum. However, what we can do instead to model the whether the current input is even or odd is to use the past outputs of the system. Thus, keeping this in mind and with a bit of trial and error, I arrived at the difference equation representing the impulse response:

$$y(n) = \delta(n) + y(n-2)$$

Therefore, substituting in $\delta(n) = x(n)$, we get the difference equation of the system to be:

$$y(n) = x(n) + y(n-2)$$

1.b) Find the difference equation representation of the system with impulse response $h_2(n)$.

Solution:

Since we know that the impulse response is $y(n)$ given when the input of the system is given by $x(n) = \delta(n)$, we can simply substitute each occurrence of $\delta(n)$ with $x(n)$ to arrive at the difference equation:

$$y(n) = x(n) - x(n-1) + x(n-2) - x(n-3)$$

1.c) Find the (A, B, C, D) representation of the system corresponding to $h_1(n)$.

Solution:

(I) Find $\boldsymbol{S(n)}$.

Let:	Then:
$s_1(n) = y(n-2)$	$s_1(n+1) = y(n-1) = s_2(n)$
$s_2(n) = y(n-1)$	$s_2(n+1) = y(n) = x(n) + y(n-2) = x(n) + s_1(n)$

Therefore, let $\boldsymbol{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \end{pmatrix}$

(II) Find $\boldsymbol{S(n+1)}$.

$$\begin{aligned} \boldsymbol{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ x(n) + s_1(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \boldsymbol{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, the $\boldsymbol{A}$ and $\boldsymbol{B}$ matrices of our system are given by:

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\boldsymbol{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

respectively.

(III) Find $y(n)$.

$$\begin{aligned} y(n) &= x(n) + y(n-2) \\ &= x(n) + s(1) \\ &= \begin{pmatrix} 1 & 0 \end{pmatrix} \boldsymbol{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, the $\boldsymbol{C}$ and $\boldsymbol{D}$ matrices of our system are given by:

$$\boldsymbol{C} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$
$$\boldsymbol{D} = \begin{pmatrix} 1 \end{pmatrix}$$

respectively.

1.d) Find the (A, B, C, D) representation of the system with impulse response $h_2(n)$.

Solution:

(I) Find $\boldsymbol{S(n)}$.

Let:	Then:
$s_1(n) = x(n-3)$	$s_1(n+1) = x(n-2) = s_2(n)$
$s_2(n) = x(n-2)$	$s_2(n+1) = x(n-1) = s_3(n)$
$s_3(n) = x(n-1)$	$s_3(n+1) = x(n)$

Therefore, let $\boldsymbol{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \\ s_3(n) \end{pmatrix}$

(II) Find $\boldsymbol{S(n+1)}$.

$$\begin{aligned} \boldsymbol{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \\ s_3(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ s_3(n) \\ x(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \boldsymbol{S(n)} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, the $\boldsymbol{A}$ and $\boldsymbol{B}$ matrices of our system are given by:

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\boldsymbol{B} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

respectively.

(III) Find $y(n)$.

$$\begin{aligned} y(n) &= x(n) - x(n-1) + x(n+2) - x(n-3) \\ &= x(n) - s_3(n) + s_2(n) - s_1(n) \\ &= \begin{pmatrix} -1 & 1 & -1 \end{pmatrix} \boldsymbol{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, the $\boldsymbol{C}$ and $\boldsymbol{D}$ matrices of our system are given by:

$$\boldsymbol{C} = \begin{pmatrix} 1 & -1 & 1 \end{pmatrix}$$
$$\boldsymbol{D} = \begin{pmatrix} 1 \end{pmatrix}$$

respectively.

2) The Frequency Response

Compute the frequency response for each of the following systems.

2.a) $y(n) = -2x(n-1) + x(n-2)$

Solution:

Since the frequency response, $H(\omega)$, of a system is given by $y(n) = H(\omega)e^{i\omega n}$ when $x(n) = e^{i\omega n}$, we can simply substitute $y(n)$ and $x(n)$ for each of these values respectively to find that:

$$\begin{aligned}
H(\omega)e^{i\omega n} &= -2e^{i\omega(n-1)} + e^{i\omega(n-2)} \\
H(\omega)e^{i\omega n} &= -2\left(e^{i\omega n} \cdot e^{-i\omega}\right) + \left(e^{i\omega n} \cdot e^{-2i\omega}\right) \\
H(\omega) &= -2e^{-i\omega} + e^{-2i\omega}
\end{aligned}$$

2.b) $y(n) = \frac{1}{2}y(n-1) + x(n)$

Solution:

Likewise, we can solve this problem with the same substitution, noting that $y(n-1) = H(\omega)e^{i\omega(n-1)}$.

$$\begin{aligned}
H(\omega)e^{i\omega n} &= \frac{1}{2}H(\omega)e^{i\omega(n-1)} + e^{i\omega n} \\
H(\omega)e^{i\omega n} &= \frac{1}{2}H(\omega)\left(e^{i\omega n} \cdot e^{-i\omega}\right) + e^{i\omega n} \\
H(\omega) &= \frac{1}{2}H(\omega)e^{-i\omega} + 1 \\
H(\omega) - \frac{1}{2}H(\omega)e^{-i\omega} &= 1 \\
H(\omega)(1 - \frac{1}{2}e^{-i\omega}) &= 1 \\
H(\omega) &= \frac{1}{1 - \frac{1}{2}e^{-i\omega}}
\end{aligned}$$

2.c) $h(n) = \delta(n-2)$

Solution:

Clearly, we can see that the difference equation representation of the system corresponding to this impulse response is given by $y(n) = x(n-2)$, thus:

$$\begin{aligned} H(\omega)e^{i\omega n} &= e^{i\omega(n-2)} \\ H(\omega)e^{i\omega n} &= e^{i\omega n} \cdot e^{-2i\omega} \\ H(\omega) &= e^{-2i\omega} \end{aligned}$$

3) Frequency and Periodicity

Given the two signals:

$$x_1(t) = \sin\left(\frac{3}{2}t\right)$$
$$x_2(n) = \cos\left(\frac{\pi}{4}n\right)$$

3.a) Give the period, frequency, and normalized frequency of x_1 (with units)

Solution:

(I) Find the period of $x_1(t)$.

Since we know that $\sin(t)$ is 2π-periodic, we can easily deduce that the period of $\sin\left(\frac{3}{2}t\right)$ must be $\frac{2\pi}{\frac{3}{2}} = \frac{4\pi}{3}$. Thus, the period of $x_1(t)$ is $\frac{4\pi}{3}$ seconds/cycle, noting that $x_1(t)$ is continuous.

(II) Find the frequency of $x_1(t)$.

The frequency of the signal is given simply as the coefficient of t inside the sin term. Thus, the frequency of the system is $\frac{3}{2}$ radians per period.

(III) Find the normalized frequency of $x_1(t)$.

The normalized frequency of a signal is given as the reciprocal of its period. Thus, the frequency of the system is $\frac{1}{\frac{4\pi}{3}} = \frac{3\pi}{4}$ seconds per period.

3.b) Give the frequency and normalized frequency of $x_2(n)$ (with units).

Solution:

(I) Find the frequency of $x_2(n)$.

The frequency of the signal is given simply as the coefficient of n inside the cos term. Thus, the frequency of the system is $\frac{\pi}{4}$ radians per step.

(II) Find the normalized frequency of $x_2(n)$.

To find the normalized frequency of $x_2(n)$, we must first find its period. To find its period, we must find the smallest possible $P \in \mathbb{N}$ such that $\forall n \in \mathbb{N}, x_2(n) = x_2(n + P)$. Therefore:

$$\cos\left(\frac{\pi}{4}n\right) = \cos\left(\frac{\pi}{4}(n+P)\right)$$
$$\cos\left(\frac{\pi}{4}n\right) = \cos\left(\frac{\pi}{4}n + \frac{\pi}{4}P\right)$$

Therefore, to set these two equal, we need $\frac{\pi}{4}P$ to be a multiple of 2π. Thus, since the smallest possible value of P which satisfies this is 8, the period of the system is given by 8 and thus, the normalized frequency of the system is given by $\frac{1}{8}$ steps per cycle.

3.c) If x_2 is played back on a computer with a sampling frequency of $f_2 = 8000\text{Hz}$, what is the pitch of the resulting audio signal?

Solution:

The pitch of a signal is given as $f_s \cdot f$, where f_s and f are the sampling frequency and normalized frequency of the signal respectively. Thus:

$$\begin{aligned} \text{pitch } &= f_s \cdot f \\ &= 8000 \cdot \frac{1}{8} \\ &= 1000 \end{aligned}$$

Thus, the pitch of the resulting audio signal is 1000 cycles/second.

4) Discrete Fourier Series

Compute the discrete Fourier series of the 4-periodic signal:

$$1, 1, 0, 0, 1, 1, 0, 0, \ldots$$

Solution:

Using the formula for the discrete formula series $\left(X_k = \frac{1}{P}\sum_{n=0}^{P-1} x(n)e^{-i\omega_0 nk}\right)$, and noting that $\omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}$, we can compute the Fourier series' coefficients as:

$$\begin{aligned}
X_0 &= \tfrac{1}{4}(1+1+0+0) &&= \tfrac{1}{2} \\
X_1 &= \tfrac{1}{4}(1+e^{-i\frac{\pi}{2}}) &&= \tfrac{1-i}{4} \\
X_2 &= \tfrac{1}{4}(1+e^{-i\frac{\pi}{2}(2)}) &&= 0 \\
X_3 &= \tfrac{1}{4}(1+e^{-i\frac{\pi}{2}(3)}) &&= \tfrac{1+i}{4}
\end{aligned}$$